

Java FSD Day 1 and 2 Summary

1. Introduction

Java is a high-level, object-oriented programming language developed by **James Gosling** at Sun Microsystems. It was initially named **Oak** and was renamed **Java** in 1995. In 2010, Oracle acquired Sun Microsystems. Java follows the principle '**Write Once, Run Anywhere (WORA)**', which means a Java program can run on any operating system that has a JVM.

2. Features of Java

- **Object-Oriented** – Everything in Java is treated as an object.
- **Platform Independent** – Java programs can run on any operating system using the JVM.
- **Simple** – Java is easy to learn and use compared to many other programming languages.
- **Secure** – It provides runtime checking and avoids direct pointer manipulation.
- **Robust** – Java has strong memory management and exception handling.
- **Multithreaded** – It can perform multiple tasks at the same time.
- **High Performance** – Java uses the Just-In-Time (JIT) Compiler to improve performance.
- **Distributed** – Java supports distributed applications using RMI and EJB.
- **Portable** – Java bytecode can run on any machine with a JVM.

3. Compiler and Interpreter

The **Compiler (javac)** converts the Java source code (.java) into bytecode (.class) and checks for syntax errors. The **Interpreter** reads and executes the bytecode. The **JIT Compiler** converts frequently used bytecode into native machine code to improve execution speed.

Program Execution Flow

.java Source Code → Compiler (javac) → .class Bytecode → JVM (Interpreter + JIT) → Native Machine Code → Output

4. JVM, JRE and JDK

JVM (Java Virtual Machine) executes Java bytecode and converts it into machine code.

JRE (Java Runtime Environment) consists of the JVM and the required libraries to run Java applications.

JDK (Java Development Kit) contains the JRE and development tools required to write, compile, debug and run Java programs.

5. JDK Tools

- **javac** – Compiles Java source code into bytecode.
- **jdb** – Used to debug Java programs.
- **javadoc** – Generates documentation from Java source code.

6. IDE

An IDE (Integrated Development Environment) helps developers write, compile, debug and manage Java projects. Popular IDEs include Eclipse, IntelliJ IDEA, NetBeans and Visual Studio Code.

7. Java Magic Bytecode

Every Java **.class** file starts with the magic number **CAFEBABE**. The JVM checks this value to verify that the file is a valid Java bytecode file before execution.

8. RMI and EJB

RMI (Remote Method Invocation) allows a Java application to call methods of another Java object running on a different system over a network.

EJB (Enterprise JavaBeans) is a server-side technology used to build secure, scalable and distributed enterprise applications.

Handwritten Class Notes

Frame Works
Spring Boot
& React

Top - down
Bottom - Up

DAY- 1

Dude's
James Gosling
@Sun Microsystems
2025
JANUARY
Sunday 05

- What is Language
- Overview of Programming Lang. ASCII key
- Types of Programming Lang.
- OOPS - To call objects in necessary places.
- JAVA - *High level Lang.
 - * Object oriented Lang.
 - * Platform Independent.

Why JAVA??

Regular Apps
uses which lang
JAVA (a) Python

• Timeline -

- 1991 - James Gosling called Oak
- 1995 - Renamed from Oak to Java
- 1996 - JDK 1.0
- 1999 - Java 2 [~~JSE~~ J2SE, J2EE]
- 2006 - Open Source [GPL - General Pub. License]
- 2010 - Oracle acquired Sun Microsystems
- 2014 - Growth once new update

HTML - Skeleton
CSS - Design Beauty
Back-end - Life



- Libraries - Many Inbuilt libs.
- Secure and Reliable

What is RMI
& EJB



Regn. No. 2924 of 2000

DAY - 2

2025
JANUARY

Tuesday 07

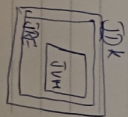
Compiler & Interpreter:

→ Interacts between Java source code & bytecode
→ [Java to .file]
→ checks syntax.

→ Live by live translation & check of the code.
→ bytecode to machine code.
→ WORA - Write Once, Run Anywhere.
→ Works with the JIT (Just in time Compiler)

JVM - ^{above} runs ~~below~~

- Runs Java Programs
- bytecodes to machine code.
- JVM interprets the bytecodes for the underlying OS.
- Work of JVM:



Loads code
Verifies code
Executes code

JRE

- JRE is the package that provides everything needed to run a JVM Program.
- built in class [java.util, java.io]
- Just as to need to run the file
- not the really stress developer.



2025

JANUARY

Thursday 09

JDK

- Java Development Kit.

It has

- ~~7~~ Tools to write, compile, debug and run

- Tools - Compiler (javac) - converts (.java) to byte code
Debugger (jdb) - find & fix
Doc (javadoc) - generate docs from code.

- Write -
Compile -

IDE - Integrated Development Environment.

- IDEs for Java,
(i) Eclipse
(ii) VS Code
(iii) Net Beans

- Features: (i) Code Editor -
(ii) Compiler -
(iii) Debugger -
(iv) Project Management - Organise files, lib
(v) Other Tools - Build tools, Git



Regn. No. 2924 of 2000

2025
JANUARY

Saturday 11

Java's Magic Bytecode: [CAFEBABE]

- Secret Signature at the start of every Java .class file
- To verify that it is a Java Byte code by JVM

JIT - Just In Time Compiler & Interpreter:

- It uses the faster native version instead of interpreting again.

Works,

- Java code (.java) $\rightarrow$ Bytecode (.class)
- JVM uses the interpreter to execute this bytecode directly.
- Each instruction in bytecode is read and performed one by one by interpreter.

Native Machine Code:

Source code (.java) $\rightarrow$ Compiler (.java) $\rightarrow$ Bytecode (.class)



Native Machine Code $\leftarrow$ JVM (interpreter + JIT)

- It is written 0s and 1s.



Regn. No. 2924 of 2000